Minindu: UISE 2027 work log and progress

Oct 9, 2026 · @Minidu

Where things stand

As of 12 Oct: the data model is pushed (10 Oct), the related-work notes are checked (11 Oct), and Framework v2 is drafted and approved (11–12 Oct). Next is the 50-guideline check, then `checkInvariants()` on 13 Oct, the meeting pack on 14 Oct, and the supervisor meeting on 15 Oct. Submission is 13 Nov 2026 AoE, 32 days away.

Framework v2 now leads with the run-time fidelity invariants, because the related-work check showed user-chosen adaptation (Madugalla) and requirement tracing (NeuroAdaptX) are not new. Three decisions go to Madam on 15 Oct; they are explained in Questions for the 15 Oct supervisor meeting.

Progress tracker

Ten tasks are done as of 14 Oct; next is the meeting pack for Friday 16 Oct, then the profile screens once wireframes arrive. Change a status by clicking its chip.

#	Task	Status	Due	Done when
1	Data model types (`forecast.ts` , `profile.ts` , `explanation.ts`)	Done	9 Oct	Files in the repo (done 9 Oct)
2	Related-work notes: Gemini run on NeuroAdaptX, Tielman 2024, Madugalla	Done	7 Oct (late)	Done 9 Oct
3	Commit and push: data model, synthetic README	Done	10 Oct	On `main` as commit 7e65fa9 (done 10 Oct)
4	Data model follow-ups	In progress	10 Oct	Supun's `forecasts.json` matches `ForecastFile` (not pushed yet); language decision settled (G121, see Blockers)
5	Related-work follow-ups: check flagged items on the page	Done	14 Oct	Done 11 Oct; save `related-work-notes.md` in `docs/research/`
6	Framework v2	Done	10–11 Oct	Done 11 Oct; Framework v2 and `Framework_v2.pdf`
6b	Replace Task 5 in the Research Approach with Framework v2	Done	14 Oct	Done 12 Oct: shared doc updated; `Research_Approach_v3.2_12Oct.pdf` and what changed
7	Check the 50 guidelines against Framework v2	Done	14 Oct	Every row in the check doc checked. Done 13 Oct: 10 covered, 21 partly, 9 UI only, 10 not covered; ratings in XAI_Framework_v2_Rating_Final.xlsx

#	Task	Status	Due	Done when
9	`model/invariants.ts`: implement `checkInvariants()`	Done	13 Oct (due 16 Oct)	Passes on `howSureToday`; fails with the right missing/wrong list on broken copies; L4 banned-phrase list included. Done 14 Oct: on main (PR #14), 10 of 10 checks pass
10	`contexts/ExplanationProfileContext.tsx` and `useExplanationProfile()`	Done	13 Oct (due 16 Oct)	Saves and loads with AsyncStorage like `UserPreferencesContext`; survives app restart. Done 14 Oct (typechecked; wired to screens with S01–S05)
11	Supervisor meeting pack	In progress	14 Oct	Questions doc done 12 Oct; add status page, guideline totals and Supun's agreement count
12	Walkthrough tasks T1–T5 step counts to Imansha	Not started	Date to set	Sent from the Start Plan doc
13	Screens S01–S06 and `ProfileQuestion`	Not started	29 Oct	Meets the build plan's "Done when" list
14	Both functions working end to end on a phone	Not started	29 Oct (freeze)	Walkthrough can start 30 Oct
15	Paper: framework section, introduction, related work	Not started	Date to set (before 13 Nov)	Drafts in the paper repo, within the page plan
16	Final check of claims and references before submission	Not started	13 Nov	No sentence makes a claim from the "cannot make" list; every reference opened

Tasks 9 and 10 were merged to main on 14 Oct. Imansha's second rating is not tracked here; his guide is Second rating of the XAI comparison table.

How to do each next task

Do them in this order; each list is the steps to finish that task.

3. Push your files (10 Oct, 10 minutes)

1. In the repo folder, run `git status`. You should see `ews-mobile/model/` (5 files), `ews-mobile/data/synthetic/README.md`, and the Research Approach v3.1 files.
2. Run `npm run typecheck` inside `ews-mobile`. It should finish with no errors.
3. Commit on the `minindu` branch, push, and open a pull request to `main`, the same way as PR #12.
4. Check on GitHub that `ews-mobile/model/forecast.ts` is there. As of 10 Oct it is not; the last commit on `main` is from 6 Oct.

4. Data model follow-ups (10–11 Oct, 30 minutes)

1. After the push, send Supun the message from chat (10 Oct). It asks him to make `forecasts.json` and `explain.ts` follow your three files.
2. When Supun pushes `forecasts.json`, tell Claude. Claude checks it against the `Forecast` type and lists anything that does not match.
3. Answer the three decisions in the next section. If you accept the recommendations, no code changes.

The three data model decisions (recommendations)

1. **Alert mode (G124):** keep all five facts, in one short line, e.g. "Flooding possible here on Thursday. Get your go-bag ready today." Recommended, because the check is the paper's main contribution and should never be switched off. No code change.
2. **Preset fill-in values:** keep them as drafted, and ask Imansha to confirm them when he sends his profile questions. No code change now.
3. **Language (G121):** use the app's existing language setting, not a new profile field. Language is not an explanation preference. No code change.

6. Framework v2 (done 11 Oct) and 6b. Put it in the Research Approach (14 Oct)

1. Framework v2 is in its doc and `Framework_v2.pdf`.
2. After the 15 Oct meeting, apply Madam's answers to the three questions.
3. Then replace Task 5 (pages 19–21) of the Research Approach with the Framework v2 text and figure, and save it as v3.2.

7. Check the 50 guidelines (12–14 Oct, about 2 hours)

1. Open How to check the 50 guidelines against Framework v2. Claude has filled in a first pass for all 50 rows.
2. Follow its 10 steps: check each row against the catalogue and the code, keep or change the Status chip, and write "OK" or your change in Your check.
3. Decide each of the six gaps at the end, and write the totals.
4. Tell Claude; the totals go into the meeting pack and the code comments get the missing guideline numbers.

5. Related-work follow-ups: check five things in the PDFs (about 1 hour)

Gemini's output is a list of leads, not checked facts. Open each PDF and confirm these five things; write the page number beside each.

1. **Tielman PDF:** use Ctrl+F to find our four quotes: "key to inclusive", "more than just simplicity", "throughout the development", "feasible". Also read the author list on page 1: is the fifth author Siebert or van der Aa?
2. **NeuroAdaptX p. 215:** read the checklist paragraph. Confirm that it is done once, before deployment, and that it says nothing about uncertainty.
3. **Madugalla Section 7.1:** read where the four-day fire forecast is described. Does it explain why the forecast says what it does? If not, "shown but not explained" is safe to write.
4. **Madugalla "cognitive impairments":** find the page (Gemini said both 12 and 15). Cite arXiv:2511.12856v2, since that is the version you read.
5. **Send the five answers to Claude.** Claude drafts the three related-work paragraphs from them, and you edit them in your own words.

9–10. Invariants check and profile store (13 Oct)

1. In `model/invariants.ts`, implement `checkInvariants(e, f)`: slot text empty → missing; slot value differs from the forecast → wrong; `e.forecastId !== f.id` → wrong.
2. For Layer 4 also check the text: the likelihood text contains the CAP word, and no banned negated phrase appears (G16).
3. Test it on `howSureToday` (must pass) and on copies with a slot removed or a value changed (must fail).
4. Copy the pattern of `contexts/UserPreferencesContext.tsx` for `ExplanationProfileContext.tsx`: one AsyncStorage key, load on start, save on change, `DEFAULT_PROFILE` until set.

11. Supervisor meeting pack (14 Oct)

1. `Framework_v2.pdf`.
2. `Supervisor_Questions_15Oct.pdf`: the invariants explained and the three questions (done 12 Oct).
3. One status page from this tracker, with the 50-guideline totals.
4. Supun's one-slide XAI comparison with the agreement count (N of 24) from 12 Oct.

Blockers and open questions

As of 12 Oct, nothing blocks the guideline check or `checkInvariants()`; the open items are Supun's `forecasts.json`, one language decision, and three questions for Madam.

Blocker or question	Who unblocks it	Needed by	Meanwhile
Supun's `forecasts.json` in the agreed shape (not on GitHub as of 12 Oct)	Supun	13 Oct	The fixture `__fixtures__/hanwella.ts` stands in
Supun's code (`supun-ml-workstream.zip`) not yet seen or pushed	Supun	13 Oct	Nothing of yours waits on it
Imansha's second rating and the comparison with Supun	Imansha, Supun	12 Oct	Needed only for the meeting pack
Wireframes for S01–S06	Imansha	22 Oct	Build `ProfileQuestion` and the profile store, which need no wireframe
Imansha's current profile questions and personas	Imansha	Before screens	Use the seven questions in Research Approach Task 3
Language (G121): the catalogue says "the language comes from the profile"; on 9 Oct we chose the app's setting	You decide	14 Oct	Recommended: keep the app setting and change the catalogue line
Q1: lead with invariants, or present both contributions equally?	Madam	15 Oct meeting	Framework v2 leads with invariants
Q2: alert mode keeps all five facts, or only hazard, place and action (G124)?	Madam	15 Oct meeting	`explanation.ts` requires all five
Q3: test the AI wording check on prepared rewordings, or build a live model?	Madam	15 Oct meeting	Plan: prepared rewordings
Keep the old condition picker beside the new profile, or drop it?	Madam	15 Oct meeting	Both exist side by side
Does a second person classify the 128 guidelines, for agreement?	Madam / team	15 Oct meeting	Your triage stands alone

Blocker or question	Who unblocks it	Needed by	Meanwhile
Where to host the online appendix	Team	Before 13 Nov	Repo `docs/` folder

Related work: what the Gemini checks found (9 Oct)

The data table is the reliable output; the report missed whole sources and should not be used. Treat every page number below as a lead until you open the page: Gemini gave different pages for the same Madugalla quote (p. 12 in one file, p. 15 in another).

Paper	What the checks support (page as Gemini gave it)	What to fix or check
NeuroAdaptX (Bunde 2026, DESRIST, LNCS 16605, pp. 205–225)	Adapts structure, modality and density (p. 205–206); self-reported ADHD, autism (ASC) and dyslexia profiles (p. 205–206); 10 design requirements DR1–DR10 from cognitive load, cognitive fit, multimedia learning and W3C COGA (p. 206, 210–212); stress-prediction task in digital wellbeing; online vignette experiment, N = 216, 36 per profile × condition (p. 217); comprehension 0.81 vs 0.73, g = 0.39, p = .004 (p. 220–221); equivalence checked before deployment with a coverage checklist of four units: pattern detected, main drivers, causal interpretation, actionable takeaway (p. 215); templated explanations, not live model output (p. 222–223)	Our claim of "no requirement-to-capability mapping" is wrong: DRs are traced to design principles and features (p. 210–215). Uncertainty: "Not stated", so our "no uncertainty" claim needs your own read. Its checklist includes "causal interpretation", a contrast with our rule that feature importance is never cause

Paper	What the checks support (page as Gemini gave it)	What to fix or check
Tielman et al. (2024), Technology in Society 79, 102685	Conceptual roadmap, no empirical evaluation (p. 2, 7–8); extends the Explanation Ontology with 2 levels, 4 modes, 6 types, 5 content elements, 5 language levels, 7 user types (p. 3–4); its scenarios do communicate uncertainty (p. 4–5); limits: no datasets for this group, hard recruitment, simplification vs accuracy (p. 4, 6–7)	11 Oct: all quotes checked on the page by Minindu (pp. 8–10). Quote 3 actually begins "Users with diverse cognitive disabilities", so fix the catalogue wording. Fifth author is Luciano Cavalcante Siebert (p. 1); Imansha's "van der Aa L" is wrong
Madugalla et al., arXiv:2511.12856v2	Four groups (p. 1); 62 guidelines → 67 requirements, 17 functional and 50 non-functional (p. 1, 7); 6 interviews (2 older adults, 4 rural) and 8 walkthrough evaluators, 2 per persona (p. 1, 9); 56 of 67 requirements addressed, 47 positively validated (p. 10); cognitive impairments outside scope (Section 10.3, p. 12; confirmed 10 Oct); adaptivity must remain explainable and controllable (p. 1, 11); Australian bushfire prototype with a four-day fire danger forecast (Section 7.1)	"Forecast shown but not explained" confirmed by Claude in arXiv v2 Section 7.1 (10 Oct): it shows the Country Fire Authority's ratings and names that source, but does not explain them. Users choose their own settings at setup and in settings (p. 11), so "user-chosen" is not new. SSRN version not uploaded, so the year question is still open

What this means for our gap statement. Two parts are now shared with prior work: user-chosen adaptation (Madugalla) and tracing requirements to features (NeuroAdaptX). Three parts remain ours and none of the sources weakens them:

1. The object explained is a safety-critical ML prediction, for people with cognitive impairments (Madugalla excludes them; NeuroAdaptX is low-stakes).

2. Fidelity is checked at run time on every adapted explanation, including AI wording, against named invariants; NeuroAdaptX checks once, by hand, before deployment.
3. Uncertainty is one of the invariants, so no profile can drop it.

Suggested one-line differentiation for the paper: "Unlike NeuroAdaptX, which checks informational equivalence once at design time on fixed templates, we check five invariants, including uncertainty, at run time on every adapted explanation of a safety-critical prediction." Confirm the NeuroAdaptX side on p. 215 before using it.

Work log

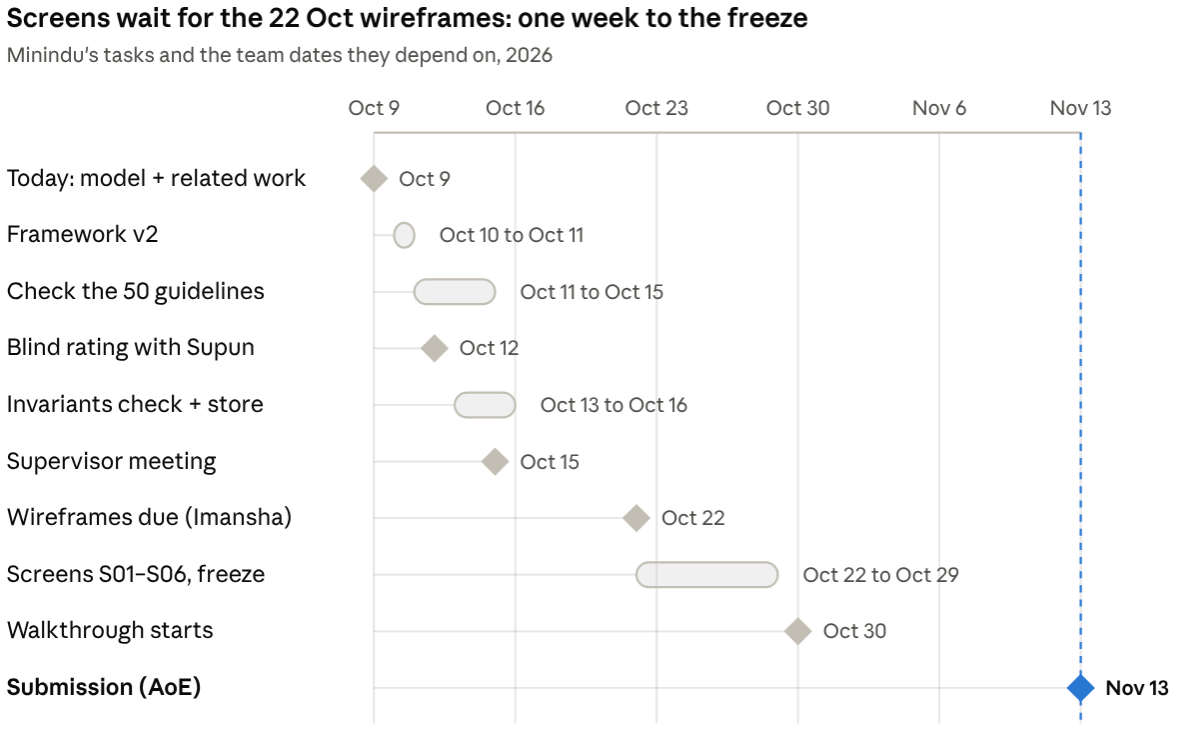
Everything finished so far, newest first; teammates' work is listed where yours depends on it.

Date	Who	What	Where it lives
14 Oct	Minindu (with Claude Code)	checkInvariants() with banned-phrase and idiom check (10 of 10 tests pass) and the profile store; coverage now 13 covered, 20 partly, 9 UI only, 8 not covered	main, PR #14 (commit 5b2a086); XAI_Framework_v2_Guideline_Coverage.xlsx
13 Oct	Minindu + Claude	50-guideline final ratings (second rating, LLM re-audit, code check); Research Approach v3.2 and its change summary; meeting pack explainer	Coverage workbook; Research_Approach_v3.2_12Oct.pdf; Research_Approach_v3.2_Changes.pdf
12 Oct	Minindu + Claude	50-guideline check prepared: first pass for all 50 rows (22 covered, 14 partly, 9 UI only, 5 not covered)	How to check the 50 guidelines

Date	Who	What	Where it lives
12 Oct	Minindu + Claude	Supervisor questions explained, with Hanwella examples	Questions for 15 Oct; `Supervisor_Questions_15Oct.pdf`
11 Oct	Minindu + Claude	Framework v2: invariants lead, alert-mode fix, check after every layer, capability table linked to code	Framework v2; `Framework_v2.pdf`
11 Oct	Minindu	Related-work facts checked on the page (Tielman quotes and author, NeuroAdaptX p. 215, Madugalla Section 7.1)	`related-work-notes.md`
10 Oct	Minindu	Data model and synthetic README pushed to `main`	Commit 7e65fa9
9 Oct	Minindu	Gemini notebook run on the three papers	`Inclusive_XAI_and_RE_Research_Papers.xlsx`, `madugalla-version-analysis.docx` (chat uploads)
9 Oct	Minindu + Claude	Data model drafted: `forecast.ts`, `profile.ts`, `explanation.ts`, Hanwella fixture	`ews-mobile/model/`
9 Oct	Minindu + Claude	Second-rating guide for Imansha	Second rating guide

Date	Who	What	Where it lives
9 Oct	Minindu + Claude	Gemini Notebook setup and prompts	Related work via Gemini Notebook
7 Oct	Minindu	Research Approach v3.1	Shared doc; `Research_Approach_v3.1_7Oct.pdf`
7 Oct	Minindu	"Updates for Supun" doc	Updates for Supun
6 Oct	Minindu	CAP 1.2 §3.2.2 certainty words verified; NeuroAdaptX and Madugalla read	`task1-leftover-checks.md`
6 Oct	Supun	SHAP, LIME, DiCE on synthetic data (8,000 rows, AUC 0.852, p = 0.291); 24-cell table, first pass	His zip (not yet seen)
5 Oct	Minindu	XAI guideline evidence catalogue v2: 50 of 128 guidelines; 38 Direct, 9 Indirect, 3 none found	`docs/research/xai-guideline-evidence-catalogue.pdf`
5 Oct	Minindu	Prototype build plan	`docs/prototype/build-plan.md`
Early Oct	Imansha	User-side literature review: 15 pages, about 100 cited	`Imansha - Literature Review Complete.pdf`
Before 5 Oct	Minindu	Repo set up and cleaned; Start Plan doc; read Tielman 2024	Start Plan

Key dates to submission



key dates to submission · 9 Oct to 13 Nov 2026

The tight stretch is 22–29 Oct: screens can only start when wireframes arrive, so build everything that needs no wireframe before then. Later dates: notification 11 Dec 2026, camera-ready 29 Jan 2027.

Rules and claims we cannot make

Reviewers will check these; every task above must respect them.

Rules:

- Adapt the explanation, never the prediction or its risk estimate.
- Profiles are by need, never diagnosis; no condition name anywhere in the app or the profile names.
- Mark every requirement as stated by a source or inferred by us; call the three profiles hypothetical.
- Label synthetic values in the app and the paper; never invent evidence or data sources.
- Never present feature importance or supporting observations as cause.
- The 5% Unlikely/Possible cut-off and Minor Flood Level = SEVERE are our own choices; say so.
- Walkthrough findings are expert findings only; no user study has been run.

Claims we cannot make:

- That the profiles improve comprehension for people with cognitive impairments.
- That words plus picture is best for them.
- That "Possible" means 30% in CAP.
- That any profile was validated.
- New from 9 Oct: that we are the first to trace requirements to features, or to let users choose their adaptation.

Links and files

- Repo: MininduBimsara/indigenous-study. Key files: `ews-mobile/model/`, `ews-mobile/data/synthetic/README.md`, `docs/prototype/build-plan.md`, `docs/research/`
- Framework v2
- Questions for the 15 Oct supervisor meeting
- How to check the 50 guidelines against Framework v2
- Second rating guide for Imansha
- Related work via Gemini Notebook
- Start Plan · Updates for Supun
- Project handoffs: `claude/handoff-2026-10-09.md` (current), `claude/handoff-2026-10-06.md`